

```
from google.colab import files
uploaded = files.upload()
```

student\_data.csv

**student\_data.csv**(text/csv) - 41983 bytes, last modified: 6/25/2026 - 100% done  
Saving student\_data.csv to student\_data.csv

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use('default')
```

```
df = pd.read_csv('/content/student_data.csv',
                 sep=None,
                 engine='python')

df.head()
```

|   | school | sex | age | address | famsize | Pstatus | Medu | Fedu | Mjob    | Fjob     | ... | famrel | freetime |
|---|--------|-----|-----|---------|---------|---------|------|------|---------|----------|-----|--------|----------|
| 0 | GP     | F   | 18  | U       | GT3     | A       | 4    | 4    | at_home | teacher  | ... | 4      | 3        |
| 1 | GP     | F   | 17  | U       | GT3     | T       | 1    | 1    | at_home | other    | ... | 5      | 3        |
| 2 | GP     | F   | 15  | U       | LE3     | T       | 1    | 1    | at_home | other    | ... | 4      | 3        |
| 3 | GP     | F   | 15  | U       | GT3     | T       | 4    | 2    | health  | services | ... | 3      | 2        |
| 4 | GP     | F   | 16  | U       | GT3     | T       | 3    | 3    | other   | other    | ... | 4      | 3        |

5 rows × 33 columns

```
print("Rows and Columns:", df.shape)
```

Rows and Columns: (395, 33)

```
df.columns.tolist()
```

```
['school',
 'sex',
 'age',
 'address',
 'famsize',
 'Pstatus',
 'Medu',
 'Fedu',
 'Mjob',
 'Fjob',
 'reason',
 'guardian',
 'traveltime',
 'studytime',
 'failures',
 'schoolsup',
 'famsup',
 'paid',
 'activities',
 'nursery',
 'higher',
 'internet',
 'romantic',
 'famrel',
 'freetime',
```

```
'goout',
'Dalc',
'Walc',
'health',
'absences',
'G1',
'G2',
'G3']
```

```
df.columns.tolist()
```

```
['school',
'sex',
'age',
'address',
'famsize',
'Pstatus',
'Medu',
'Fedu',
'Mjob',
'Fjob',
'reason',
'guardian',
'travelttime',
'studytime',
'failures',
'schoolsup',
'famsup',
'paid',
'activities',
'nursery',
'higher',
'internet',
'romantic',
'famrel',
'freetime',
'goout',
'Dalc',
'Walc',
'health',
'absences',
'G1',
'G2',
'G3']
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 395 entries, 0 to 394
Data columns (total 33 columns):
#   Column          Non-Null Count  Dtype
---  -
0   school          395 non-null    object
1   sex              395 non-null    object
2   age              395 non-null    int64
3   address          395 non-null    object
4   famsize          395 non-null    object
5   Pstatus          395 non-null    object
6   Medu             395 non-null    int64
7   Fedu             395 non-null    int64
8   Mjob             395 non-null    object
9   Fjob             395 non-null    object
10  reason           395 non-null    object
11  guardian         395 non-null    object
12  travelttime      395 non-null    int64
13  studytime        395 non-null    int64
14  failures         395 non-null    int64
15  schoolsup        395 non-null    object
16  famsup           395 non-null    object
17  paid             395 non-null    object
18  activities       395 non-null    object
19  nursery          395 non-null    object
```

```
20 higher      395 non-null object
21 internet    395 non-null object
22 romantic    395 non-null object
23 famrel      395 non-null int64
24 freetime    395 non-null int64
25 goout       395 non-null int64
26 Dalc        395 non-null int64
27 Walc        395 non-null int64
28 health      395 non-null int64
29 absences    395 non-null int64
30 G1          395 non-null int64
31 G2          395 non-null int64
32 G3          395 non-null int64
dtypes: int64(16), object(17)
memory usage: 102.0+ KB
```

```
df.isnull().sum()
```

|                   | 0 |
|-------------------|---|
| <b>school</b>     | 0 |
| <b>sex</b>        | 0 |
| <b>age</b>        | 0 |
| <b>address</b>    | 0 |
| <b>famsize</b>    | 0 |
| <b>Pstatus</b>    | 0 |
| <b>Medu</b>       | 0 |
| <b>Fedu</b>       | 0 |
| <b>Mjob</b>       | 0 |
| <b>Fjob</b>       | 0 |
| <b>reason</b>     | 0 |
| <b>guardian</b>   | 0 |
| <b>traveltime</b> | 0 |
| <b>studytime</b>  | 0 |
| <b>failures</b>   | 0 |
| <b>schoolsup</b>  | 0 |
| <b>famsup</b>     | 0 |
| <b>paid</b>       | 0 |
| <b>activities</b> | 0 |
| <b>nursery</b>    | 0 |
| <b>higher</b>     | 0 |
| <b>internet</b>   | 0 |
| <b>romantic</b>   | 0 |
| <b>famrel</b>     | 0 |
| <b>freetime</b>   | 0 |
| <b>goout</b>      | 0 |
| <b>Dalc</b>       | 0 |
| <b>Walc</b>       | 0 |
| <b>health</b>     | 0 |
| <b>absences</b>   | 0 |
| <b>G1</b>         | 0 |
| <b>G2</b>         | 0 |
| <b>G3</b>         | 0 |

**dtype:** int64

```
print("Duplicates:", df.duplicated().sum())
```

```
df.drop_duplicates(inplace=True)
```

Duplicates: 0

```
df.describe()
```

|              | age        | Medu       | Fedu       | traveltime | studytime  | failures   | famrel     | freetime   |
|--------------|------------|------------|------------|------------|------------|------------|------------|------------|
| <b>count</b> | 395.000000 | 395.000000 | 395.000000 | 395.000000 | 395.000000 | 395.000000 | 395.000000 | 395.000000 |
| <b>mean</b>  | 16.696203  | 2.749367   | 2.521519   | 1.448101   | 2.035443   | 0.334177   | 3.944304   | 3.235443   |
| <b>std</b>   | 1.276043   | 1.094735   | 1.088201   | 0.697505   | 0.839240   | 0.743651   | 0.896659   | 0.998862   |
| <b>min</b>   | 15.000000  | 0.000000   | 0.000000   | 1.000000   | 1.000000   | 0.000000   | 1.000000   | 1.000000   |
| <b>25%</b>   | 16.000000  | 2.000000   | 2.000000   | 1.000000   | 1.000000   | 0.000000   | 4.000000   | 3.000000   |
| <b>50%</b>   | 17.000000  | 3.000000   | 2.000000   | 1.000000   | 2.000000   | 0.000000   | 4.000000   | 3.000000   |
| <b>75%</b>   | 18.000000  | 4.000000   | 3.000000   | 2.000000   | 2.000000   | 0.000000   | 5.000000   | 4.000000   |
| <b>max</b>   | 22.000000  | 4.000000   | 4.000000   | 4.000000   | 4.000000   | 3.000000   | 5.000000   | 5.000000   |

```
df['G3'].describe()
```

|              | G3         |
|--------------|------------|
| <b>count</b> | 395.000000 |
| <b>mean</b>  | 10.415190  |
| <b>std</b>   | 4.581443   |
| <b>min</b>   | 0.000000   |
| <b>25%</b>   | 8.000000   |
| <b>50%</b>   | 11.000000  |
| <b>75%</b>   | 14.000000  |
| <b>max</b>   | 20.000000  |

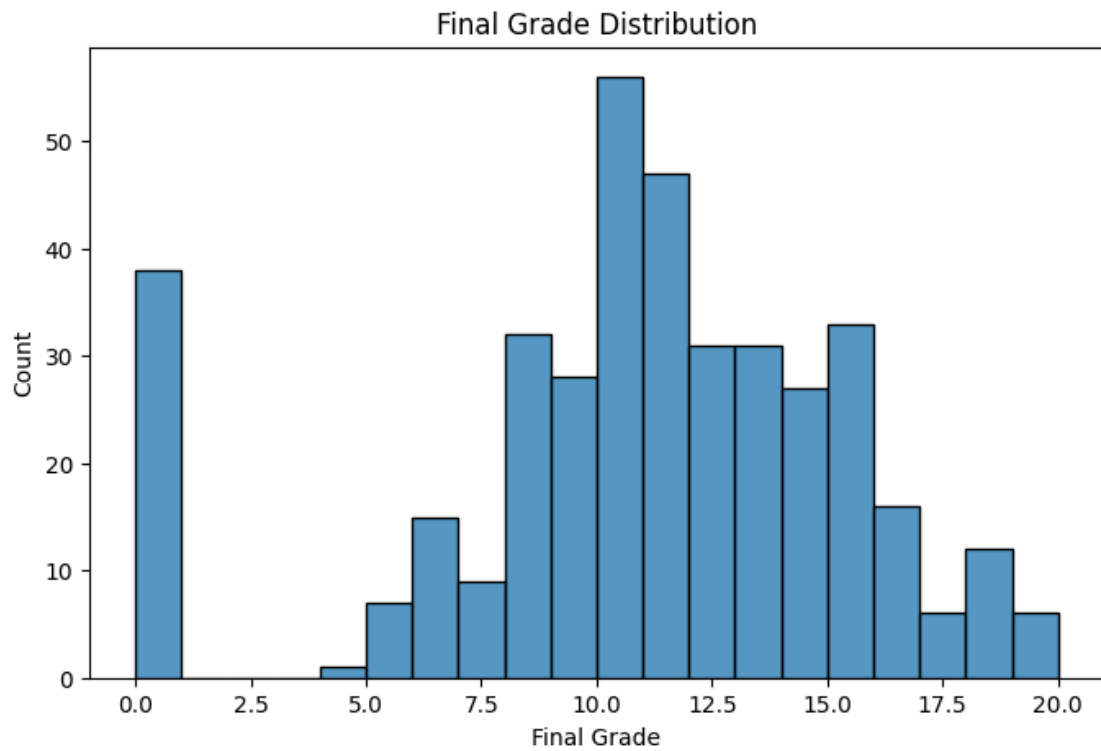
**dtype:** float64

```
plt.figure(figsize=(8,5))

sns.histplot(df['G3'], bins=20)

plt.title("Final Grade Distribution")
plt.xlabel("Final Grade")
plt.ylabel("Count")

plt.show()
```

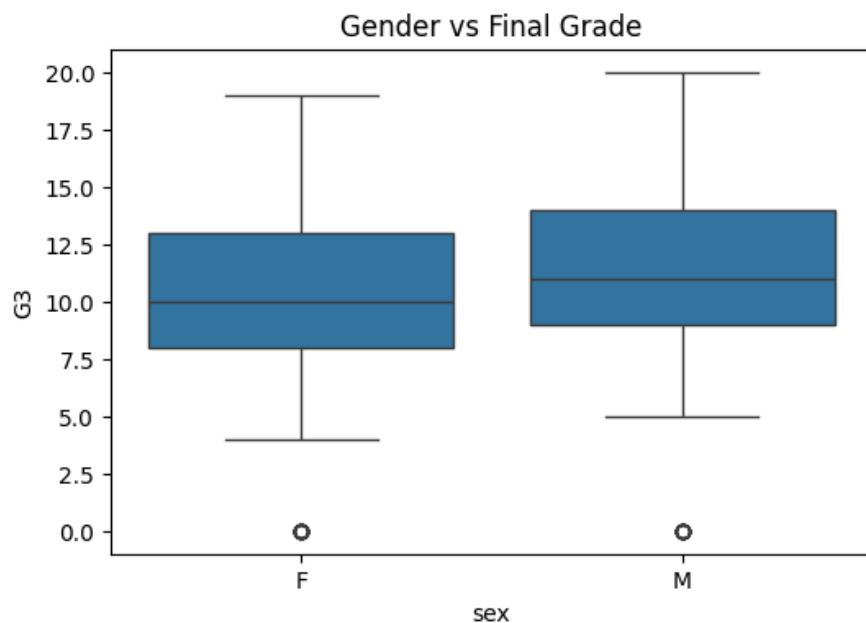


```
plt.figure(figsize=(6,4))

sns.boxplot(x='sex',
            y='G3',
            data=df)

plt.title("Gender vs Final Grade")

plt.show()
```

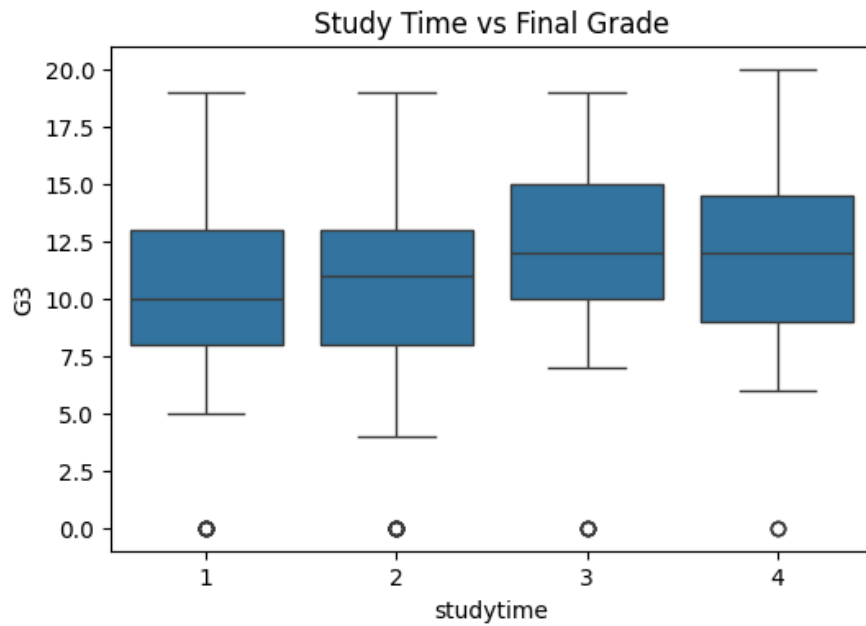


```
plt.figure(figsize=(6,4))

sns.boxplot(x='studytime',
            y='G3',
            data=df)

plt.title("Study Time vs Final Grade")
```

```
plt.show()
```

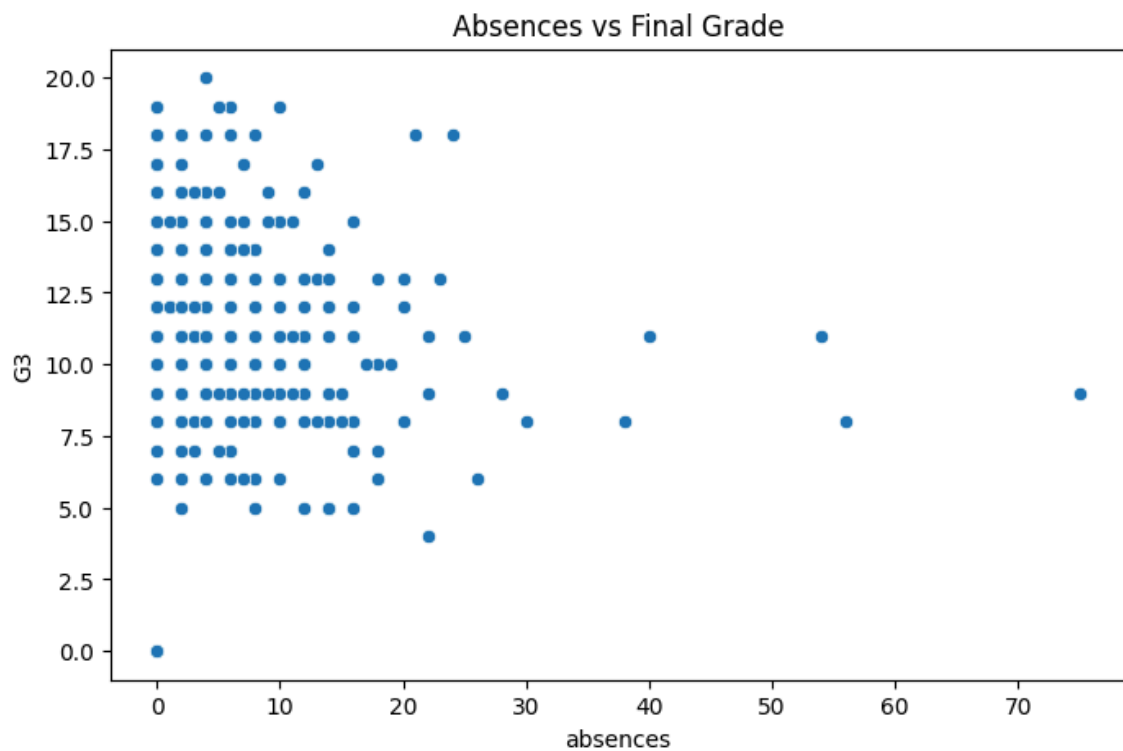


```
plt.figure(figsize=(8,5))

sns.scatterplot(x='absences',
                y='G3',
                data=df)

plt.title("Absences vs Final Grade")

plt.show()
```

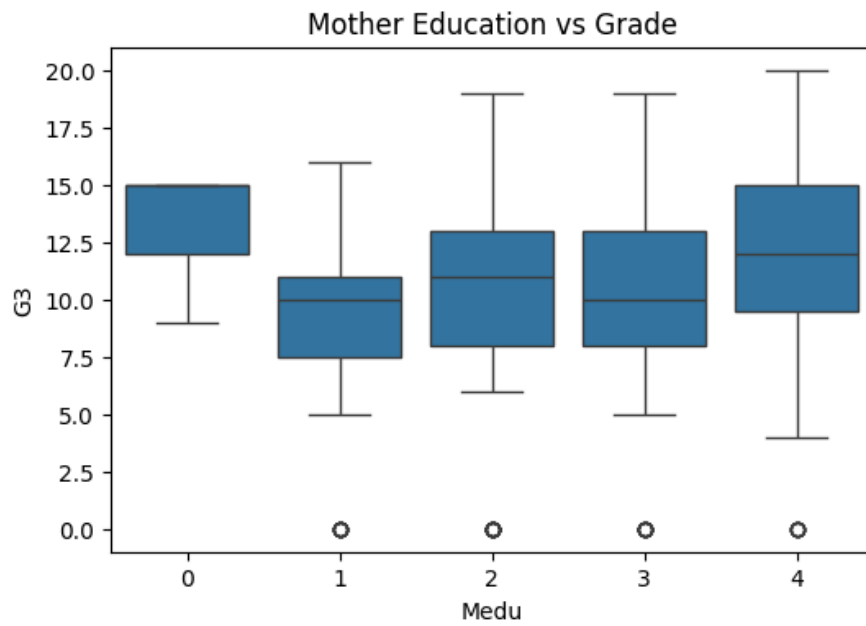


```
plt.figure(figsize=(6,4))

sns.boxplot(x='Medu',
            y='G3',
            data=df)

plt.title("Mother Education vs Grade")
```

```
plt.show()
```

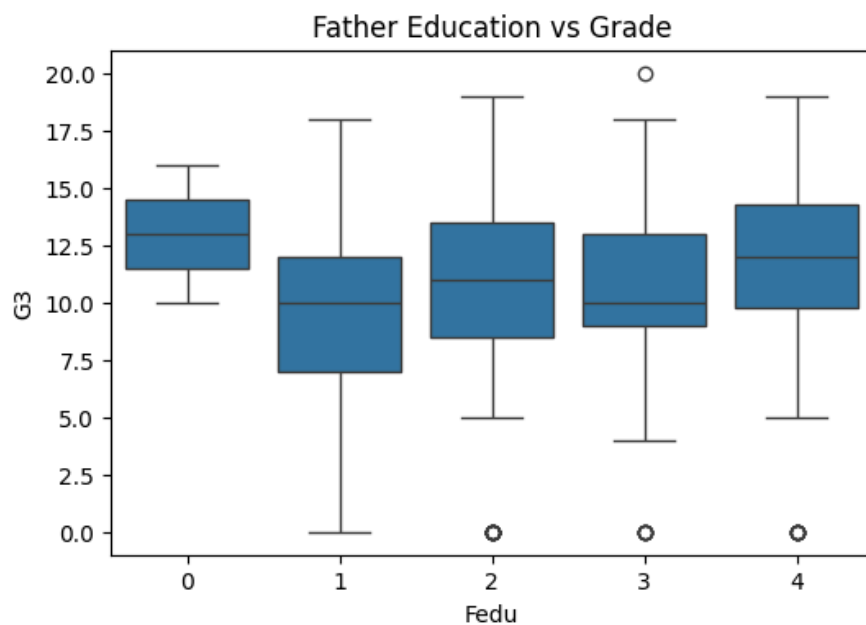


```
plt.figure(figsize=(6,4))

sns.boxplot(x='Fedu',
            y='G3',
            data=df)

plt.title("Father Education vs Grade")

plt.show()
```



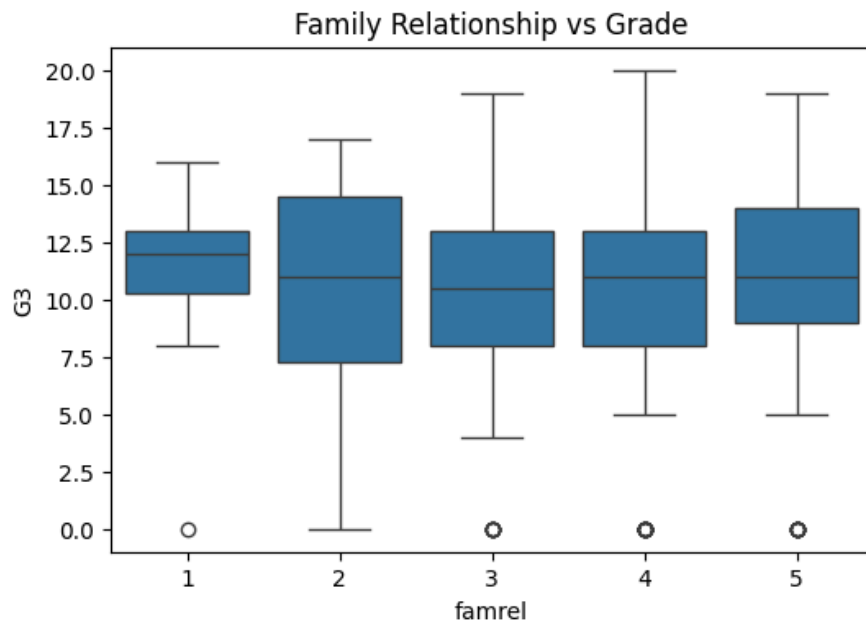
```
plt.figure(figsize=(6,4))

sns.boxplot(x='famrel',
            y='G3',
            data=df)

plt.title("Family Relationship vs Grade")

plt.show()
```





```
from sklearn.preprocessing import LabelEncoder

df_encoded = df.copy()

le = LabelEncoder()

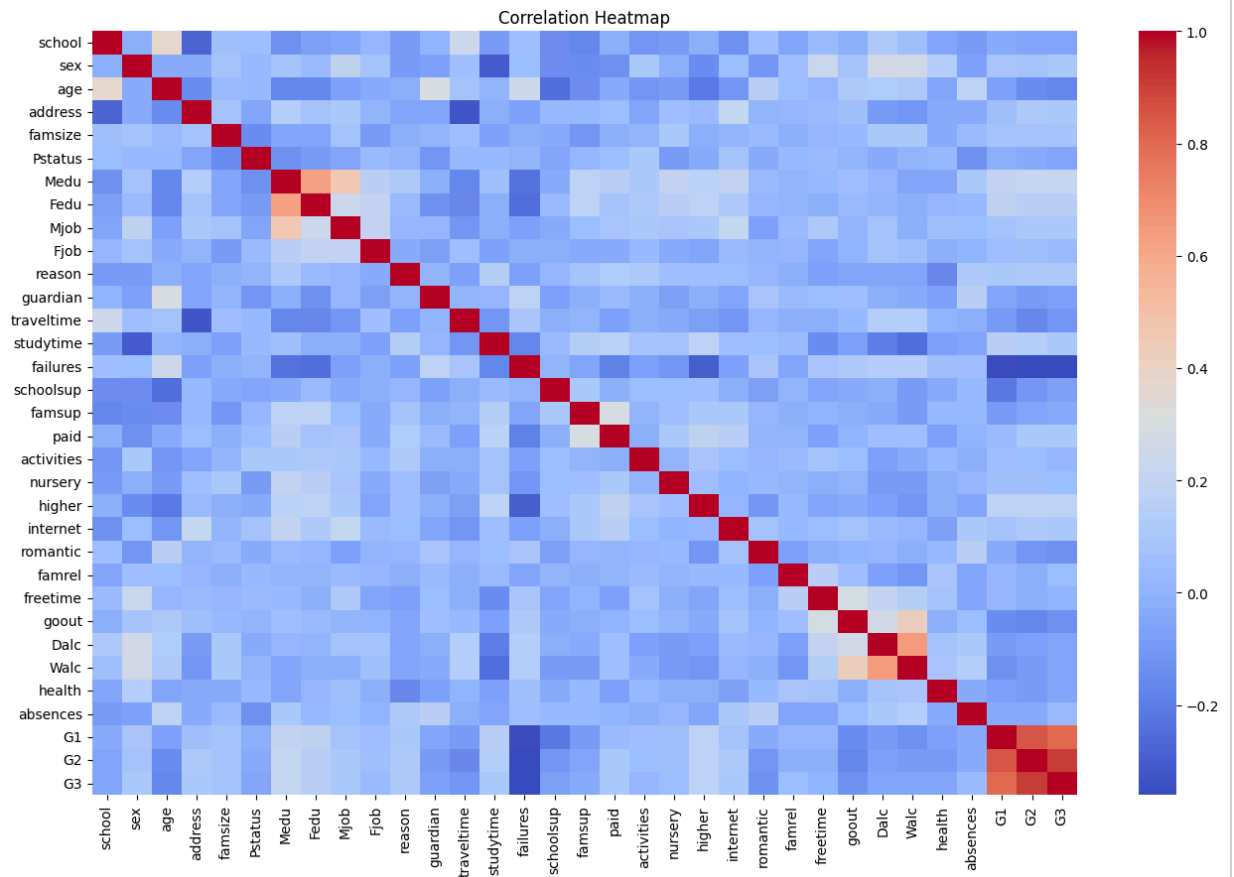
for col in df_encoded.columns:
    if df_encoded[col].dtype == 'object':
        df_encoded[col] = le.fit_transform(df_encoded[col])
```

```
plt.figure(figsize=(16,10))

sns.heatmap(df_encoded.corr(),
            cmap='coolwarm')

plt.title("Correlation Heatmap")

plt.show()
```



```
corr = df_encoded.corr()['G3']

corr.sort_values(ascending=False)
```

|                  | G3       |
|------------------|----------|
| <b>G3</b>        | 1.000000 |
| <b>G2</b>        | 0.904868 |
| <b>G1</b>        | 0.801468 |
| <b>Medu</b>      | 0.217147 |
| <b>higher</b>    | 0.182465 |
| <b>Fedu</b>      | 0.152457 |
| <b>reason</b>    | 0.121994 |
| <b>address</b>   | 0.105756 |
| <b>sex</b>       | 0.103456 |
| <b>Mjob</b>      | 0.102082 |
| <b>paid</b>      | 0.101996 |
| <b>internet</b>  | 0.098483 |
| <b>studytime</b> | 0.097820 |
| <b>famsize</b>   | 0.081407 |
| <b>nursery</b>   | 0.051568 |